

Devesh Kumar Choubey

✉ deveshchoubey4086@gmail.com | 📁 LeetCode | 🐙 GitHub

PROFESSIONAL SUMMARY

Software Engineer with expertise in Java, Python, Spring Boot, and system integration. Proven experience designing scalable backend services, REST APIs, and secure web applications with measurable performance gains. Passionate about building Generative AI and LLM-based solutions, with hands-on problem-solving demonstrated through 400+ algorithmic challenges on LeetCode. Adept at collaborating across crossfunctional teams to deliver clean, modular, and self-sustaining code aligned with enterprise engineering practices.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL,

AI/ML & Deep Learning: PyTorch, OpenAI Whisper, Transformers, LLMs, LMMs, Embeddings, Diffusion Models (familiar), Prompt Engineering, RAG

Generative AI & Agents: Spring AI, Ollama, Pinecone, Gumloop, LangChain-style agentic workflows, LLM fine-tuning, Semantic Search

Frameworks & Tools: FastAPI, Spring Boot, Hibernate, REST APIs, Git, IntelliJ IDEA, Postman

Databases: MySQL, Vector Databases (Pinecone), JDBC

Core CS: Data Structures & Algorithms, OOPS, DBMS, Operating Systems, Linear Algebra, Probability

Soft Skills: Research Mindset, Problem-Solving, Cross-functional Collaboration, Technical Writing, Ownership

EXPERIENCE

System Engineer

Infosys

Sep 2025 – Present

Bengaluru, India

- Designed **automated validation pipelines** for enterprise applications using Python and API frameworks, improving regression coverage by **40%** across critical workflows.
- Collaborated with cross-functional teams to debug, profile, and optimize integration workflows, reducing **defect leakage by 25%**.
- Engaged with internal ML/automation initiatives, applying scripting and data-driven analysis to **accelerate test cycle execution by 30%**.

PROJECTS

AI-Powered Job Portal with Semantic Search | *Python, PyTorch, Spring AI, Ollama, Pinecone, Embeddings, LLMs*

- Architected an end-to-end GenAI application serving **100+ simulated users**, leveraging LLM embeddings and a vector database (Pinecone) for resume-to-job semantic matching with **85% relevance accuracy**.
- Implemented secure JWT-based authentication and authorization using **Spring Security**, ensuring protected access to role-specific functionalities.
- Engineered a horizontally scalable Job Portal deployment using Docker Compose and NGINX load balancing across 2 Spring Boot instances, validating performance with K6 load tests of **1,000(7,400 requests)** concurrent users and identifying BCrypt authentication as the primary latency bottleneck.

Self-Evolving AI Resume Tailoring Agent | *Python, LLMs, Prompt Engineering, Gumloop, Agentic Workflows*

- Built an agentic AI workflow that autonomously analyzes job descriptions, reasons over role requirements, and generates tailored outputs, **reducing manual effort by 90%**.
- Engineered **multi-step LLM prompt pipelines** with iterative self-evaluation, improving ATS keyword alignment scores by **45%**.
- Developed a template-aware code-generation module producing **dynamic LaTeX output**, demonstrating LLM-driven code synthesis and evaluation.

Speech-to-Text Multimodal Transcription System | *Python, PyTorch, OpenAI Whisper, FastAPI*

- Built a speech-to-text application using the OpenAI Whisper model to convert audio recordings into accurate text transcripts.
- Developed FastAPI backend services to handle audio uploads, transcription requests, and transcript generation workflows
- Improved transcription quality using audio preprocessing techniques such as noise reduction and audio segmentation

EDUCATION

Nitte Meenakshi Institute of Technology

Bengaluru, India

Bachelor of Engineering in Electronics and Communication Engineering; Dec 2021 – Jul 2025

- 7.65 CGPA